

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 11****Topics covered :**

1. Introduction to Bifurcations
2. Saddle-Node, Transcritical and Pitchfork Bifurcations

- Recall that for a differential equation of the form $\frac{dy}{dt} = f(y)$, the equilibrium points y^* are the values of y for which $f(y) = 0$. However, if the parameters of the differential equation (or the underlying system) change, the differential equation itself may change, potentially altering the equilibrium points. Let μ represent the parameter that varies in the system.

For example, consider an RL circuit with a DC voltage source. If the inductance L changes (keeping in mind that the permeability of the core depends on temperature), the governing differential equation for the current response also changes. The modified equation could be represented as:

$$\frac{di}{dt} = \frac{1}{\mu} [V_{dc} - iR] \quad (1)$$

where μ is the varying parameter. In the case of the RL circuit, a variation in inductance does not change the equilibrium point. On the other hand, if the resistance in the circuit changes, the governing differential equation becomes

$$\frac{di}{dt} = \frac{1}{L} [V_{dc} - i\mu] \quad (2)$$

In this case, a change in μ (which represents the varying resistance) will cause a shift in the equilibrium point.

- **Bifurcation analysis** plays a crucial role in understanding how the equilibrium points of a differential equation (DE) change as system parameters vary. For the purposes of this course, we will assume that the system is characterized by a single parameter that undergoes variation. Additionally, we will introduce and explore different types of bifurcations through examples that illustrate the fundamental characteristics of each type. More explicitly, an autonomous differential equation with one varying parameter can be represented as

$$\frac{dy}{dt} = f(y, \mu) \quad (3)$$

The goal of bifurcation analysis [1] is to understand how the behavior of equilibrium points changes in response to variations in the parameter μ , specifically, how the system's response evolves as μ changes.

- **Saddle-Node bifurcation:** Consider a first order autonomous ODE

$$\frac{dy}{dt} = f(\mu, y) = \mu - y^2 \quad (4)$$

The equilibrium points for the autonomous ODE (4) are determined by

$$\mu - y^2 = 0 \quad (5)$$

This yields the equilibrium points

$$y^* = \pm\sqrt{\mu} \quad (6)$$

The existence and stability of equilibrium points are governed by the value of μ .

1. Case I ($\mu > 0$): For $\mu > 0$, there are two equilibrium points, $\sqrt{\mu}$ and $-\sqrt{\mu}$. The equilibrium point at $y = \sqrt{\mu}$ is stable since $\left. \frac{df}{dy} \right|_{\sqrt{\mu}} = -2\sqrt{\mu} < 0$. On the other hand, the equilibrium point at $y = -\sqrt{\mu}$ is stable since $\left. \frac{df}{dy} \right|_{-\sqrt{\mu}} = 2\sqrt{\mu} > 0$. Note that the equilibrium points move closer as the value of μ decreases.
2. Case II ($\mu = 0$): For $\mu = 0$, the system has only one equilibrium point namely, $y^* = 0$.
3. Case III ($\mu < 0$): For $\mu < 0$, the system does not have any equilibrium points.

Thus, for this ODE, the variation in μ causes the two equilibrium points to move towards each other, collide, and annihilate each other [1]. This entire process, where two equilibrium points move towards each other, collide, and then annihilate as a system parameter μ passes through a **critical value** or **bifurcation point** is the key characteristic of a **Saddle node bifurcation**. In such bifurcations, a pair of equilibrium points of opposite stability (one stable and one unstable) merge at the bifurcation point, and beyond that point, they no longer exist.

One effective way to visually represent the changes in the equilibrium points as the parameter μ varies is through a **Bifurcation diagram**. In such a diagram, the parameter μ is typically plotted along the x -axis, while the equilibrium points y^* are plotted along the y -axis. The bifurcation diagram shows how the locations of the equilibrium points change as μ is varied, and it also captures changes in their nature, including their stability (i.e., whether they are stable or unstable). In the bifurcation diagram, equilibrium points are plotted for different values of μ . The transition of **stable equilibrium points** is shown using **solid lines** while the transition of **unstable equilibrium points** is shown using **dashed lines**. The bifurcation diagram for the autonomous ODE (4) is shown in Fig. 1.

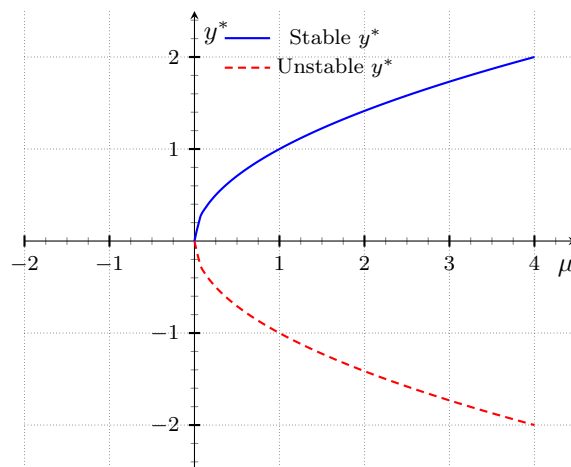


Fig. 1: Bifurcation diagram for $y' = \mu - y^2$ (Saddle-node bifurcation)

- **Transcritical bifurcation:** Consider the first-order autonomous differential equation

$$\frac{dy}{dt} = \mu y - y^2 \quad (7)$$

The equilibrium points for the ODE 7 are determined by

$$\mu y - y^2 = 0 \quad (8)$$

This yields the equilibrium points

$$y_1^* = 0 \text{ and } y_2^* = \mu \quad (9)$$

In this case, one of the equilibrium points i.e., $y^* = 0$ does not change while the other equilibrium point varies with μ . More importantly, their stability, which is determined by (10) is effected by the value of μ .

$$\left. \frac{df(\mu, y)}{dy} \right|_{y^*} = \mu - 2y^* \quad (10)$$

1. Case I ($\mu > 0$): For $\mu > 0$, $f'(y_1^*) = \mu > 0$, indicating that $y_1^* = 0$ is an unstable equilibrium. On the other hand, $f'(y_2^*) = -\mu < 0$, indicating that $y_2^* = \mu$ is a stable equilibrium.
2. Case II ($\mu < 0$): For $\mu < 0$, $f'(y_1^*) = \mu < 0$, indicating that $y_1^* = 0$ is an stable equilibrium. On the other hand, $f'(y_2^*) = -\mu > 0$, indicating that $y_2^* = \mu$ is an unstable equilibrium.
3. Case III ($\mu = 0$): The two equilibrium points coincide

Thus as $\mu \rightarrow 0$, the two equilibrium points move towards each other and at the **bifurcation point** $\mu = 0$, they exchange their stability properties. Bifurcations of this kind, wherein the two equilibrium points of opposite stability (one stable and one unstable) cross each other and exchange their stability at the bifurcation point is the key characteristic of **transcritical bifurcation**. The bifurcation diagram for the ODE (7) is shown in Fig. 2.

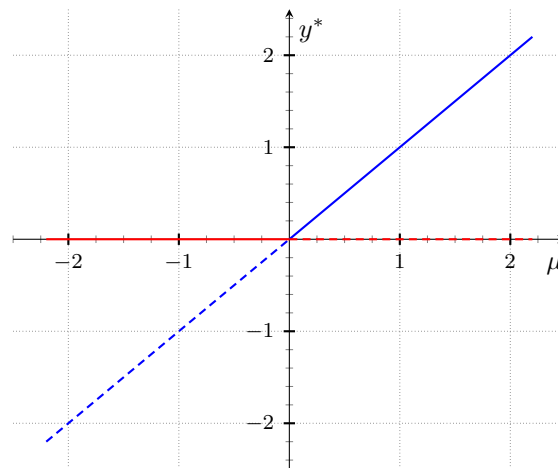


Fig. 2: Bifurcation diagram for $y' = \mu y - y^2$ (Transcritical bifurcation)

- **Pitchfork bifurcation:** Consider the first-order autonomous differential equation

$$\frac{dy}{dt} = \mu y - y^3 \quad (11)$$

The equilibrium points for the ODE 11 are determined by

$$\mu y - y^3 = 0 \quad (12)$$

This yields the equilibrium points

$$y_1^* = 0, \quad y_2^* = \sqrt{\mu}, \quad \text{and} \quad y_3^* = -\sqrt{\mu} \quad (13)$$

In this case, one of the equilibrium points i.e., y_1^* does not change while the other two equilibrium points vary with μ and only exist when $\mu > 0$. Thus, their existence as well as stability, which is determined by (14) is effected by the value of μ .

$$\left. \frac{df(\mu, y)}{dy} \right|_{y^*} = \mu - 3(y^*)^2 \quad (14)$$

1. Case I ($\mu > 0$): For $\mu > 0$, $f'(y_1^*) = \mu > 0$, indicating that $y_1^* = 0$ is an unstable equilibrium. On the other hand, $f'(y_2^*) = f'(y_3^*) = -2\mu < 0$, indicating that y_2^* and y_3^* are stable.
2. Case II ($\mu < 0$): For $\mu < 0$, there is only one equilibrium point i.e., $y_1^* = 0$ and since $f'(y_1^*) = \mu < 0$, the equilibrium point is stable.
3. Case III ($\mu = 0$): The three equilibrium points coincide

Thus when $\mu < 0$, the system has only one stable equilibrium point and as μ crosses the [bifurcation point](#) i.e., $\mu = 0$, two new symmetric stable equilibrium points are created and the other equilibrium switches its stability property. Bifurcations of this kind are called [transcritical bifurcation](#). The bifurcation diagram for the ODE (11) is shown in Fig. 3.

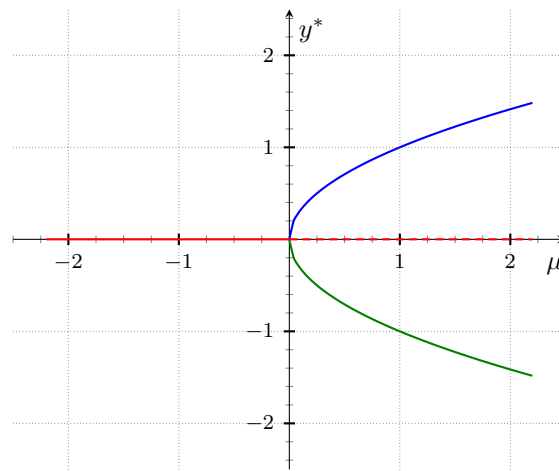


Fig. 3: Bifurcation diagram for $y' = \mu y - y^3$ (Pitchfork bifurcation)

References

- [1] S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed. Cambridge, MA: Perseus Books, 2018.